

RESEARCH ARTICLE

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Key Points:

- Trends in OH between 1980 and 2015 were modeled using observed ozone column, water vapor, methane, model-simulated NO_x, and Hadley cell width
- Most (94%) of the expected, CH₄-induced decline in OH was offset by increases due to H₂O, NO_x, overhead O₃, and tropical widening (in order of impact)
- Trends depend on time frame examined; NO_x and overhead O₃ drive most of OH increase between 1980 and 2005

Supporting Information:

- Supporting Information S1

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Changes in Global Tropospheric OH Expected as a Result of Climate Change Over the Last Several Decades

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Abstract The oxidizing capacity of the troposphere is controlled primarily by the abundance of hydroxyl radical (OH). The global mean concentration of tropospheric OH, [OH]^{TROP} (the burden of OH in the global troposphere appropriate for calculating the lifetime of methane) inferred from measurements of methyl chloroform has remained relatively constant during the past several decades despite rising levels of methane that should have led to a decline. Here we examine other factors that may have affected [OH]^{TROP} such as the changing values of stratospheric ozone, rising tropospheric H₂O, varying burden of NO_x (=NO+NO₂), rising temperatures, and widening of the climatological tropics due to expansion of the Hadley cell. Our analysis suggests the positive trends in [OH]^{TROP} due to H₂O, NO_x, and overhead O₃, and tropical expansion are large enough (Δ [OH]^{TROP} = +0.95 ± 0.18%/decade) to counter almost all of the expected decrease in [OH]^{TROP} due to rising methane (Δ [OH]^{TROP} = −1.01 ± 0.05%/decade) over the period 1980 to 2015, while variations in temperature contribute almost no trend (Δ [OH]^{TROP} = −0.02 ± 0.02%/decade) in [OH]^{TROP}. The approximated impact of Hadley cell expansion on [OH]^{TROP} is also a small but not insignificant factor partially responsible for the steadiness of tropospheric oxidizing capacity over the past several decades, which free-running models likely do not capture.

1. Introduction

A realistic representation of the atmospheric lifetime of methane (CH₄) is important for the accurate modeling of global warming. Methane is the second most important anthropogenic greenhouse gas (IPCC, 2013). The mixing ratio of tropospheric methane was nearly constant at around 1775 ppb from about 1997 to 2006. Since 2007, methane has been rising at a rate varying between 4.7 ppb/year and 12.5 ppb/year (or 0.3–0.7%/year). The cause of the recent rise in methane is debated: some ascribe the rise to emissions from natural wetlands (Schaefer et al., 2016), others to a combination of fossil fuels and wetlands (Kirschke et al., 2013), others to changes in livestock and agriculture (Saunio et al., 2017; Wolf et al., 2017), and still others to a decline in the main sink of methane, reaction with the hydroxyl radical (OH) (McNorton et al., 2016; Rigby et al., 2017; Turner et al., 2017).

Our paper is motivated by the temporal variations in the atmospheric loss of methane, which is generally assumed to be constant on the decadal time scale. The loss of methane in the atmosphere is dominated by its reaction with OH in the troposphere (Lelieveld et al., 1998). Due to its role in oxidizing methane and many other atmospheric species, we strive to understand the time-varying evolution of the global burden of tropospheric OH ([OH]^{TROP}).

Numerous modeling and chemical inversion studies, in which observations and known emissions of another species are used to quantify OH concentrations, point to gaps in our knowledge of the value and variability of [OH]^{TROP}. For example, calculations conducted using numerous global models show a 60%–80% variation in the value of the methane lifetime due to loss via OH (Fiore et al., 2009; Nicely et al., 2016, 2017; Shindell et al., 2006; Voulgarakis et al., 2013). Nicely et al. (2016) showed that the observationally constrained tropospheric column abundance of OH in the Tropical Western Pacific during winter 2014 tended to exceed the value found in global models, due to the tendency of the models to underestimate the observed abundances of O₃ and NO_x (= NO + NO₂). Numerous other studies have also shown that models systematically

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underestimate methane lifetime by 10%–15% relative to values inferred from methyl chloroform (MCF: CH_3CCl_3) inversions (i.e., models overestimate OH) (Naik et al., 2013; Prather et al., 2012; Voulgarakis et al., 2013). Additional studies using observational constraint suggest models regularly misrepresent hemispheric distributions of OH (Monks et al., 2015; Naik et al., 2013; Patra et al., 2014; Strode et al., 2015). Strode et al. (2015) found that substituting observed values of ozone and reanalysis values of water vapor for the simulated ones improved their estimate of the methane lifetime using a chemistry-climate model. López-Comí et al. (2016) used a single-column photochemical model to determine that water vapor and ozone biases within a global chemistry-climate model have the largest impact on OH over New Zealand. Finally, Nicely et al. (2017) showed that the primary causes of variations in $[\text{OH}]^{\text{TROP}}$ simulated by global models were, in order of importance, different chemical mechanisms, various treatments of the photolysis frequency of O_3 that leads to production of $\text{O}(^1\text{D})$, and modeled O_3 and CO. In general, modeling studies tend to attribute OH variations and model differences to a wide assortment of factors, highlighting the difficulty of constraining $[\text{OH}]^{\text{TROP}}$.

There are also large uncertainties regarding the temporal evolution of the global oxidizing capacity of the troposphere, over both the past few decades and extending to the end of this century. An investigation based upon chemical inversion of MCF data by Montzka et al. (2011) found that the interannual variability in $[\text{OH}]^{\text{TROP}}$ between 1998 and 2008 is considerably smaller, ~2%, than previously estimated for 1978–2004 by Prinn et al. (2005) and for 1980–2000 by Bousquet et al. (2005), both estimating variability on the order of 10%. Montzka et al. (2011) attribute the high interannual variability from previous studies to large uncertainties in the emissions of MCF and methane. The previously mentioned work of López-Comí et al. (2016) found no significant trend in OH between 1994 and 2010, based on their single column analysis in the remote southern Pacific Ocean. Two recent chemical inversion studies indicate increases in $[\text{OH}]^{\text{TROP}}$ by ~7%–10% between the 1990s and early 2000s, leading to the plateau in methane growth rate during this time, but differ in their attribution of the rise in methane burden since 2007 (Rigby et al., 2017; Turner et al., 2017). McNorton et al. (2016) likewise use MCF observations within a modeling framework to determine that variations in $[\text{OH}]^{\text{TROP}}$ largely account for the slowdown in methane growth. Finally, model representations of how $[\text{OH}]^{\text{TROP}}$ will respond to future climate change diverge; using a common scenario for future emissions of methane, some models predict an increase in the tropospheric oxidizing capacity by year 2100, whereas others forecast a decrease (Voulgarakis et al., 2013). The level of interannual variability in $[\text{OH}]^{\text{TROP}}$ and the direction of its recent and future trends remain important outstanding questions.

Given the uncertainty in how $[\text{OH}]^{\text{TROP}}$ has and will continue to respond to climate change, there is a need to understand what factors have controlled $[\text{OH}]^{\text{TROP}}$ over a period of time with sufficient observational constraints. Past studies have simulated variations in the OH burden using global atmospheric chemistry models tied to a particular observation, for example, to lightning as was done by Murray et al. (2013) and to CO reanalysis as was done by Gaubert et al. (2017). Also, Murray et al. (2014) examined the factors driving variations in the tropospheric OH burden on longer time scales, since the Last Glacial Maximum, using the GEOS-Chem chemical transport model and found that overhead O_3 , H_2O , and emissions of NO_x and reactive carbon together explain about three fourths of the computed variability of OH. Similarly, Holmes et al. (2013) evaluated $[\text{OH}]^{\text{TROP}}$ and its sensitivity to various factors over a shorter period (1997–2009) in three chemical transport models and identified temperature, H_2O , overhead O_3 , and NO_x from biomass burning and lightning as dominant influences. While the drivers of OH changes were consistent among the three models, the magnitude of interannual variability in computed methane lifetime was about a factor of 2 lower than the variability in observed MCF decay rates, though uncertainties in the two major MCF observation networks are likely too large to sufficiently constrain methane lifetime variations over the short period examined (Holmes et al., 2013). The potential for multiple factors to influence the time variations in $[\text{OH}]^{\text{TROP}}$ has been demonstrated repeatedly. However, the limited extent to which observations can be used to constrain the concentration of OH computed in a global model framework restricts how well we can expect to *know* $[\text{OH}]^{\text{TROP}}$ through the use of tools such as chemical transport models.

Our goal is to empirically model changes in $[\text{OH}]^{\text{TROP}}$ for 1980–2015 using as many observational constraints as possible, based upon first principles of OH chemistry. We extend this analysis back to 1980, close to the start of the modern satellite era, to compare our results with previously published work. Satellite observations of H_2O and column ozone (O_3), along with ground-based observations of methane are analyzed to infer

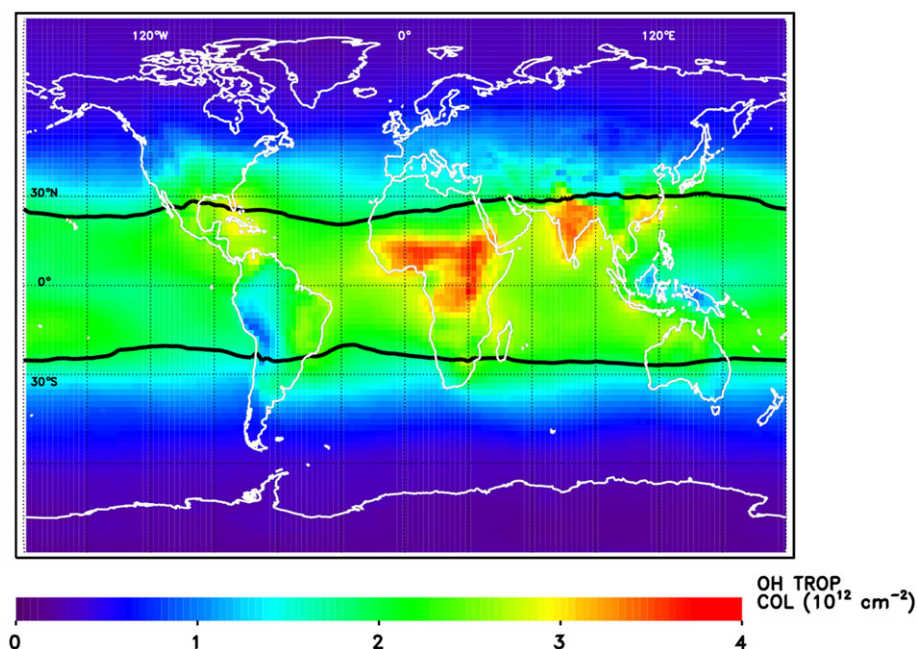
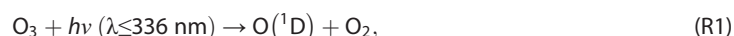


Figure 1. Annual average tropospheric OH columns are shown from the Global Modeling Initiative model for year 1980, used as a baseline in this analysis. Annual average tropical boundaries are indicated as black lines.

changes in $[\text{OH}]^{\text{TROP}}$ over the past 3.5 decades. Reanalysis fields are used to evaluate the role of H_2O prior to the availability of tropospheric observations from satellite (i.e., before 2002) as well as examine the impact of temperature changes on $[\text{OH}]^{\text{TROP}}$. With the aid of a global chemical transport model, we also include the effect of time varying emissions of NO_x ($= \text{NO} + \text{NO}_2$) on $[\text{OH}]^{\text{TROP}}$. Finally, we investigate the role that the expansion of the tropics (Seidel et al., 2008), a phenomenon not captured in most global models, has in influencing the variability of $[\text{OH}]^{\text{TROP}}$.

Primary production of OH occurs via O_3 photolysis to excited state $\text{O}(^1\text{D})$ followed by reaction with water vapor:



Because this formation pathway depends on the abundance of H_2O and the actinic flux at wavelengths $\leq 336 \text{ nm}$, the highest concentrations of OH are found in the tropical lower and middle troposphere. Stratospheric column O_3 is low in the tropics, allowing greater penetration of ultraviolet (UV) photons to the troposphere than in the extratropics. This, along with more H_2O due to higher temperatures, results in $[\text{OH}]^{\text{TROP}}$ being dominated by OH in the tropics (Figure 1).

Secondary sources of OH include recycling of HO_2 by reaction with NO:



Lelieveld et al. (2016) show that on a global basis, this source of OH is close in magnitude to primary production by $\text{O}(^1\text{D}) + \text{H}_2\text{O}$ (33% of all tropospheric OH production occurs by primary production, while 30% occurs by $\text{NO} + \text{HO}_2$). Particularly in strong source regions of NO_x , OH abundances are likely determined by this recycling pathway. Additionally, alkyl peroxy radicals (RO_2) that are generated following the oxidation of volatile organic compounds (VOCs) can either react with NO or HO_2 , with pathways leading to the regeneration of OH. To illustrate with the simplest hydrocarbon chemistry, the methane oxidation product methylperoxy radical (CH_3O_2) can react as follows:



The HO_2 produced in (R5) can then recycle to OH via (R3), while the formaldehyde (HCHO) product can either generate HO_x ($= \text{OH} + \text{HO}_2$) radicals if photolyzed or have a neutral effect following reaction with OH and subsequent production of HO_2 . Alternatively, CH_3O_2 can react with HO_2 as follows:



This reaction pathway is more common in the background atmosphere, where concentrations of NO_x are lower (Jacob et al., 2005; Müller et al., 2016). Similar to HCHO, the methylhydroperoxide product (CH_3OOH) either produces HO_x or has no influence on HO_x , depending on whether loss of CH_3OOH occurs by photolysis or by reaction with OH.

Loss of HO_x radicals is controlled mainly by the self-reaction of HO_2 , (R7), especially in the low- NO_x environment that dominates the tropics where the bulk of the tropospheric OH abundance exists.



The hydrogen peroxide (H_2O_2) product is highly soluble and readily removed from the atmosphere through wet deposition. In areas with elevated NO_x , OH can also be removed through reaction with NO_2 to form another soluble product, nitric acid (HNO_3):



However, relatively low- NO_x conditions prevail throughout the bulk of the troposphere, meaning that the self-reaction of HO_2 (R7) is the dominant sink for HO_x radicals (Jaeglé et al., 2001). Throughout this study, we consider the chemistry of OH in the context of the HO_x family due to our interest in the mean oxidation state of the troposphere and how it evolves over time. If one were interested in the fast chemistry of the OH radical over short time scales, a strict budgeting of OH loss by its reactions with a host of VOC species and its recycling via a wide range of HO_2 and RO_2 reactions would be required. Considering the HO_x radical group as a whole, though, allows us to simplify the key chemistry to its primary production ((R1) followed by (R2)) and loss pathways ((R7) and (R8)).

While the chemistry detailed here illustrates the simplest secondary production pathways of OH, Lelieveld et al. (2016) posit the importance of recycling OH from more complex oxygenated VOCs (OVOCs) in continental, low- NO_x regimes, such as the Amazon basin. The chemistry of OVOCs is represented in our baseline model, and the effect of OVOCs on OH described by Lelieveld et al. (2016) is expected to be limited to a very small altitude range (i.e., 1–2 km) over tropical forests, because of the short lifetime of the compounds that drive this chemistry. Due to the lack of observational constraints on temporal variations of OVOCs (Kim et al., 2015; Read et al., 2012) and knowledge gaps concerning the recycling pathway of OVOCs (Brewer et al., 2017; Millet et al., 2010; Müller et al., 2016; Naik et al., 2010), we have not included this term in our assessment of factors that govern changes in the oxidizing capacity of the global troposphere.

Evidence from the past two decades, based on various diagnostics of tropical extent, suggests the climatological tropics are expanding. This phenomenon can also be viewed as a strengthening or widening of the Hadley circulation, since the location of descent is generally regarded as the boundary between tropics and extratropics. Numerous studies estimating the rate of tropical widening are summarized by Seidel et al. (2008). They include rates of widening determined by analyzing tropopause heights (Birner, 2010; Lu et al., 2009; Seidel & Randel, 2007), subtropical jet stream locations (Archer & Caldeira, 2008; Fu et al., 2006; Fu & Lin, 2011; Hu & Fu, 2007), meridional mass streamfunctions (Hu & Fu, 2007; Mitas & Clement, 2005), outgoing longwave radiation (Hu & Fu, 2007), and total O_3 columns (Hudson, 2012; Hudson et al., 2006; Follette-Cook et al., 2009).

The findings of observational studies of tropical widening are often presented with inconsistent context, for example, reporting rates of widening representative of one hemisphere versus both hemispheres. To simplify

interpretation of the literature, we summarize early analyses estimated rates of expansion of the tropical belt, globally, as high as $3.0^\circ/\text{decade}$ (Seidel & Randel, 2007); however, refinement of methods and continued observation have resulted in lower estimates (Allen et al., 2014; Birner, 2010; Davis & Rosenlof, 2012). We emphasize that all of these studies are based on observations or reanalysis data sets. The most comprehensive analyses to date are Davis and Rosenlof (2012), who reported that the meridional mass streamfunction shows a robust 1.0° to $1.5^\circ/\text{decade}$ global rate of widening, while other diagnostics suggest insignificant trends, and Allen et al. (2014), who calculated rates of $0.35 \pm 0.09^\circ/\text{decade}$ in the Northern Hemisphere (NH) and $0.17 \pm 0.10^\circ/\text{decade}$ in the Southern Hemisphere (SH), as well as $0.52 \pm 0.13^\circ/\text{decade}$ globally (all based on observations). Despite the latest evidence that these rates are lower than first suggested, concerns persist due to the inability of global models to simulate the extent of tropical widening witnessed to date (Allen et al., 2014; Johanson & Fu, 2009). In general, while many models do simulate a widening of the Hadley cell in response to climate change, nearly all models considerably underestimate the magnitude of the observed widening (Allen et al., 2014). Here we quantify the effect of tropical widening on $[\text{OH}]^{\text{TROP}}$, which has yet to be considered in the literature.

The mechanism through which widening of the tropics likely affects OH chemistry is through the exposure of a larger portion of the troposphere to regions with low amounts of overhead column O_3 . Total column O_3 is lowest in the tropics due to the upward motion of the Brewer-Dobson overturning circulation (Seidel et al., 2008). This large-scale flow pattern carries stratospheric O_3 from the tropics poleward, resulting in a buildup of O_3 , and hence higher columns, in the extratropics. Indeed, the boundary between high and low column O_3 air masses can be used to distinguish the extratropics from the tropics (Hudson et al., 2006). As with the Hadley circulation in the troposphere, modeling studies find that increasing greenhouse gas concentrations will result in strengthening of the Brewer-Dobson circulation (Butchart et al., 2006; Garcia & Randel, 2008; Li et al., 2010), so it logically follows that the region of low column O_3 will expand in conjunction with the climatological tropics. This phenomenon has been observed in measurements of total column O_3 (Hudson, 2012; Hudson et al., 2006). In the regions that transition from extratropical to tropical classification, lower column O_3 will increase UV photon flux into the troposphere, likely contributing to higher values of $[\text{OH}]^{\text{TROP}}$ and thus reduced lifetime of methane. Further discussion of the mechanisms that we hypothesize are driving changes in $[\text{OH}]^{\text{TROP}}$ is provided in section 2.5.6.

In this study, we empirically model the changes in $[\text{OH}]^{\text{TROP}}$ that occurred from 1980–2015 by considering global observations of overhead O_3 , H_2O , methane, and temperature as well as the simulated effects of varying NO_x emissions and tropical expansion. The effect of each component on $[\text{OH}]^{\text{TROP}}$ is evaluated based on fundamental principles of atmospheric chemistry. Results of this analysis quantify which contributors to variations in $[\text{OH}]^{\text{TROP}}$ are most important to accurately model past and future long-term trends in the oxidizing capacity of the troposphere.

2. Methods

2.1. GMI Model

We use a 3-D field of OH calculated by the Global Modeling Initiative (GMI) chemical transport model (Duncan et al., 2007; Strahan et al., 2007) as the starting scenario for the distribution and burden of $[\text{OH}]^{\text{TROP}}$. Output for year 1980 is taken from a longer historical simulation conducted by GMI for the Chemistry-Climate Model Initiative (CCMI; Eyring et al., 2013; Morgenstern et al., 2017). Specifically, this output is taken from the Ref-C1SD historical specified dynamics simulation covering years 1980–2016. Monthly mean OH concentrations are output on a 2.5° longitude $\times$ 2° latitude grid, with 72 vertical layers from the surface to 0.015 hPa. While it may be desirable to initialize our OH field with a comprehensive set of box model simulations constrained by observations, the many species and parameters that influence OH chemistry are not measured sufficiently at the global scale to enable such a determination of $[\text{OH}]^{\text{TROP}}$. Instead, we rely on the GMI chemical transport model, which in turn utilizes realistic meteorological parameters and emissions of trace gases. GMI is constrained to meteorological fields from the Modern-Era Retrospective Analysis for Research and Applications Version 2 (MERRA-2) reanalysis data set (Gelaro et al., 2017) and run with a troposphere-stratosphere chemical mechanism that includes 117 species, 322 thermal reactions, and 81 photolysis reactions (Duncan et al., 2007; Strahan et al., 2007). For notes on availability of this and all forthcoming data sets, see the Acknowledgments section.

Anthropogenic and biomass burning emissions for the CCMI simulation come from the MACCity inventory (Granier et al., 2011). The MACCity inventory linearly interpolates the decadal Atmospheric Chemistry and Climate Model Intercomparison Project emissions (Lamarque et al., 2010) to each year on a sector-by-sector basis until 2000. MACCity anthropogenic emissions for 2000 to 2010 are based on the Representative Concentration Pathway 8.5. Biomass burning emissions are based on a combination of the Reanalysis of the Tropospheric chemical composition over the past 40 years (RETRO) inventory (Schultz et al., 2008), which spans the period 1960–2000, and the Global Fire Emissions Database version 2 (GFED2) (van der Werf et al., 2006) for 1997–2008, with the two data sets harmonized to avoid an unphysical jump (Lamarque et al., 2010). While Lamarque et al. (2010) only provide decadal monthly mean biomass burning emissions, resulting in a lack of interannual variability, Granier et al. (2011) calculate monthly emissions on an annual basis for the MACCity inventory using the RETRO and GFED2 carbon emission data sets. The GFED2 inventory uses burned area and fire count data from the Moderate-Resolution Imaging Spectroradiometer (MODIS) instrument on board the Terra satellite to generate biomass burning emissions. As such, a model constrained to these emissions could be expected to reasonably reproduce variations in NO_x that have occurred due to biomass burning. Lightning NO_x (LNO_x) is calculated online within GMI following Allen et al. (2010) and varies intra-annually, though annual emissions are constrained to 6.5 Tg N/year. GMI also includes an online calculation of NO_x emissions from soil, which vary depending on temperature and precipitation following Yienger and Levy II (1995).

2.2. DSMACC Box Model

We use the Dynamically Simple Model of Atmospheric Chemical Complexity (DSMACC; Emmerson & Evans, 2009) box model to calculate the sensitivity of OH to various chemical and physical parameters. The DSMACC model is configured to use the Master Chemical Mechanism (Jenkin et al., 1997; Saunders et al., 2003) version 3.3.1 (Jenkin et al., 2015). Inputs to the box model include pressure, latitude, longitude, temperature, CO , O_3 , H_2O , CH_4 , NO , NO_2 , and overhead O_3 column, which are taken from the GMI model simulation. Instead of conducting box model simulations for entire 3-D GMI fields, we bin the output by variables dependent on the sensitivity being examined and by latitude and pressure. For example, the sensitivity of OH to changes in methane is found as a function of both methane and NO_x , the latter determining whether OH is produced or consumed as a result of VOC oxidation. For each sensitivity, box model simulations examine variations in both a primary variable (e.g., methane) and a secondary variable (e.g., NO_x) that we identify as modifying the response of OH to the primary variable. The secondary variable is selected based on cursory box model analysis of which inputs drive the largest variability in $\delta [\text{OH}]/\delta [\text{primary variable}]$. While this approach accounts for some level of coupling between the processes influencing OH, it does not fully capture all indirect effects resulting from modification of the primary variable. To do so would require a full 3-D global model simulation with interactive chemistry; we explore this further with the GMI chemical transport model in section 3.

For all sensitivity calculations, we divide output from GMI into six latitude bins (90°S – 60°S , 60°S – 30°S , 30°S – 0° , 0° – 30°N , 30°N – 60°N , and 60°N – 90°N) and four pressure bins (surface–900 hPa, 900–750, 750–300, and 300–tropopause). Then, for the example of calculating the OH sensitivity to methane, NO_x concentrations from GMI that fall within a given latitude/pressure bin are further divided to give 10 NO_x bins. Since methane is the independent variable being examined, we divide the full range of the observational record into 10 methane bins in order to capture nonlinear responses. All other box model constraints listed above are computed as the averages of GMI tropospheric values that occur within a specified latitude and pressure bin. For each simulation, the box model is run to diurnal steady state and the final 24 hr of calculated concentrations are averaged to find daily mean values, which are directly applicable to our monthly mean-based analysis of $[\text{OH}]^{\text{TROP}}$. Simulations conducted for 10 methane bins, 10 NO_x bins, 6 latitude bins, and 4 pressure bins results in a total of 2,400 box model simulations for each sensitivity. That is, an entire set of 2,400 simulations is conducted with a focus on understanding the response of OH to a perturbation in a single variable. This is performed on a monthly basis, for 1 month per season (January, April, July, and October) to minimize computation time. Box model sensitivity calculations are performed for five parameters: overhead O_3 , H_2O , methane, temperature, and NO_x . The total number of box model simulations, as a result, is 48,000. While this set of box model computations is large in number, it is still much smaller than if we attempted to calculate individual changes in OH resulting from perturbations of the

five observed parameters listed above, for the conditions unique to each global model grid box. As such, our approach strikes a balance between computational efficiency and a thorough exploration of the chemical phase space that applies to the major reactions affecting concentrations of OH. Details of the other sensitivity calculations are provided in section 2.5.

2.3. Calculation of $[\text{OH}]^{\text{TROP}}$

The calculation of $[\text{OH}]^{\text{TROP}}$ requires a mass weighting to account for the diminishing area of higher-latitude bands and the decreasing density of air at higher altitudes. In addition, a reactivity weighting is used to find a value of $[\text{OH}]^{\text{TROP}}$ appropriate for the oxidation of methane. We use the methane mass and reaction rate weightings that were suggested by Lawrence et al. (2001):

$$[\text{OH}]^{\text{TROP}} = \frac{\sum([\text{OH}] \times k_{\text{OH}+\text{CH}_4}(T) \times M_{\text{CH}_4})}{\sum(k_{\text{OH}+\text{CH}_4}(T) \times M_{\text{CH}_4})} \quad (1)$$

Here $[\text{OH}]$, T , and M_{CH_4} are monthly mean values of OH number density, temperature, and mass of methane in each GMI model grid box, respectively. Summations cover all latitudes, longitudes, and pressure levels within the troposphere. Tropopause altitudes are standard model output, based on a blended definition that includes vertical thermal gradients and potential vorticity (Wilcox et al., 2012).

2.4. Observational Data

Measurements of total column O_3 are obtained from the National Aeronautics and Space Administration (NASA) Solar Backscatter Ultraviolet (SBUV) Merged Ozone Data Set (MOD), version 8.6 (Frith et al., 2014). This data set combines measurements from the SBUV series of ultraviolet radiometers on board several satellites, the Total Ozone Mapping Spectrometer instruments that flew onboard several satellites, and the Ozone Monitoring Instrument onboard the Aura satellite. This and all observational data sets were up to date at the time of paper submission, January 2018. Column O_3 values are provided as zonal monthly means in 5° latitude bins. Temporal variations in total column O_3 anomalies relative to 1980 for polar, midlatitude, and tropical latitude bins in each hemisphere are shown in Figure S1 in the supporting information.

Water vapor data are from the Atmospheric Infrared Sounder (AIRS) instrument onboard the NASA Aqua satellite. We use the Level 3, version 6 monthly product, which provides H_2O mixing ratios on a $1^\circ \times 1^\circ$ latitude/longitude grid at 12 pressure levels ranging from 1,000 to 100 hPa. However, since data from this satellite instrument are only available for September 2002 onward, we also analyze H_2O fields from the MERRA-2 reanalysis data set (Molod et al., 2015) to extend back to 1980. MERRA-2 monthly mean specific humidity values are extracted on a 0.5° latitude $\times$ 0.625° longitude grid for 72 pressure levels from the surface to 0.015 hPa. Gelaro et al. (2017) showed good agreement between specific humidity from MERRA-2 and from radiosonde observations, particularly in the tropical lower troposphere. The MERRA-2 data set begins in 1979, so full coverage of our study period is attained. Temperature values are also obtained from the MERRA-2 reanalysis data set.

Methane data used here are from the National Oceanic and Atmospheric Administration Cooperative Global Air Sampling Network, updated from that described by Dlugokencky et al. (2009). Surface, globally averaged mixing ratios of methane are provided as monthly means from data collected weekly at 46 remote sites starting in July 1983. To extend this data set back to 1980, we calculate the average trend from 1983 to 1990, a period over which the methane growth rate was relatively stable, and linearly interpolate backward.

The NO_x emissions within the GMI model are compared to satellite observations of fires to evaluate how well the model captures interannual variations due to biomass burning. This is performed as a check of internal consistency, since the emissions inventory used as input to GMI is itself constrained to MODIS fire data. Our purpose in showing this is to establish a connection to observations for our evaluation of NO_x . For this, fire count data from the MODIS instrument on board the Terra satellite are analyzed. Specifically, we use the Level 3 fire pixel data version 5, provided on a monthly basis at 0.5° resolution, and corrected for cloud cover (Giglio et al., 2016). These data are available for years 2001 onward. Further details of this analysis are presented in section 2.5.5.

2.5. Calculation of the OH Response

We start with the base 3-D field of OH concentrations obtained from the GMI model. The value of $[\text{OH}]^{\text{TROP}}$ calculated from this field for each month serves as our initial burden, which we assume to be representative of 1980. We then calculate perturbations to this OH field using OH sensitivities found with the DSMACC box model and observed changes in the sensitivity parameters. For example, combining the OH sensitivity to methane with the observed change in methane yields the change in OH due to methane. In other words, the box model-calculated sensitivity is multiplied offline by the change in the variable being examined. The perturbed OH field is then used to recalculate $[\text{OH}]^{\text{TROP}}$ for all years of our analysis. The difference between the perturbed $[\text{OH}]^{\text{TROP}}$ and the baseline 1980 value then represents the impact on $[\text{OH}]^{\text{TROP}}$ due to the individual factors (overhead O_3 , H_2O , methane, NO_x , and tropical widening) we examine. Below, we describe the relationships between OH and these variables used to calculate the $[\text{OH}]^{\text{TROP}}$ response.

2.5.1. OH Response to Overhead O_3

The process we use to evaluate the response of $[\text{OH}]^{\text{TROP}}$ to variations in overhead O_3 involves several distinct steps. Observations of total column O_3 from the MOD data set described in section 2.4 are used to derive observation-based variations in J (O^1D). Box model calculations are also used to find the sensitivity of $[\text{OH}]$ to changes in $[\text{O}^1\text{D}]$. We evaluate this sensitivity because it is the direct mechanism through which OH primary production is altered. First, we provide details on how the calculation of J (O^1D) using observations of total column O_3 is carried out.

Variations in $[\text{O}^1\text{D}]$ are evaluated by calculating the change in J (O^1D) caused by observed changes in total column O_3 using the photolysis mechanism Fast-JX version 7.1 (Bian & Prather, 2002; Wild et al., 2000). The 2-D field of vertical O_3 profiles from the GMI model is scaled to match the observed total column abundance of O_3 from MOD. Then, J (O^1D) values are calculated by Fast-JX for each vertical level within GMI that lies within the troposphere, using the adjusted partial overhead O_3 profile. Inputs to the Fast-JX model include a U.S. Standard Atmosphere file containing vertical profiles of temperature, geometric altitude, air density, pressure, and O_3 mixing ratios in units of ppm. The latter of these variables spans 18 latitude bins (85°S – 85°N in 10° increments), 12 months, and 31 pressure levels (1,000–0.2 hPa). We overwrite the standard O_3 mixing ratio profiles using the GMI output scaled to match observations for a given year. For each call to the Fast-JX program, we also provide latitude, longitude, Julian day, month, Greenwich Mean Time, and a value for surface albedo obtained from GMI. This is done for each year (1980–2015), each month, and, for computational efficiency, every other latitude of the GMI grid (4° increments from 90°S to 90°N) and every fourth hour of the day, zonally averaged. Our use of select latitudes and hours was determined to have practically no effect on our results, compared to use of every latitude and hour, and saved considerable computation time. A total of 119,232 (46 latitudes $\times$ 6 hr $\times$ 12 months $\times$ 36 years) Fast-JX simulations are conducted using this methodology.

Changes in J (O^1D) relative to year 1980 are assumed to be directly proportional to variations in $[\text{O}^1\text{D}]$. The impact on $[\text{OH}]$ is found by combining these changes in $[\text{O}^1\text{D}]$ with the sensitivity of OH to $[\text{O}^1\text{D}]$, or $\delta [\text{OH}]/\delta [\text{O}^1\text{D}]$. We evaluate values of $\delta [\text{OH}]/\delta [\text{O}^1\text{D}]$ using the DSMACC box model for the six latitude bins and four pressure bins described in section 2.2. Overhead O_3 column and local tropospheric $[\text{O}_3]$ values from GMI are averaged into 10 bins that span their respective ranges for each region and used as constraints within the box model simulation. We vary overhead O_3 instead of $[\text{O}^1\text{D}]$ or J (O^1D) because constraint of the latter quantities results in fixed values within the box model, despite progression of the simulation through several diel cycles until steady state is reached. Since we calculate our sensitivities using 24 hr average $[\text{OH}]$ output from each simulation, fixed values of $[\text{O}^1\text{D}]$ or J (O^1D) would yield unrealistic results. Constraints for all other box model inputs noted in section 2.2 are defined as the mean values from GMI within the specified latitude and pressure bounds. Output from the box model simulations are used to calculate $\delta [\text{OH}]/\delta [\text{O}^1\text{D}]$ as a function of local $[\text{O}_3]$, overhead O_3 , latitude, and pressure. We use $C_{\text{O}^1\text{D}}$ to denote $\delta [\text{OH}]/\delta [\text{O}^1\text{D}]$. The units of $C_{\text{O}^1\text{D}}$ are percent change of $[\text{OH}]$ divided by percent change in $[\text{O}^1\text{D}]$. The sensitivity is applied to the base 3-D OH field obtained from GMI by calculating $\delta [\text{O}^1\text{D}] \times C_{\text{O}^1\text{D}}$ on a point-by-point basis. The value of $\delta [\text{O}^1\text{D}]$ is determined using the Fast-JX photolysis mechanism, as described above, while a value of $C_{\text{O}^1\text{D}}$ appropriate to the latitude, pressure, local $[\text{O}_3]$, and overhead O_3 of a given grid cell is selected from our calculated sensitivities. The value of $\delta [\text{OH}]$ that results from this calculation is applied to $[\text{OH}]$ for

each grid point. After this perturbation in the 3-D field of OH from GMI is applied, values of $[\text{OH}]^{\text{TROP}}$ are calculated using equation (1) for each month and year.

2.5.2. OH Response to H₂O

The sensitivity of $[\text{OH}]$ to variations in $[\text{H}_2\text{O}]$ is also evaluated using the DSMACC box model. Calculation of $C_{\text{H}_2\text{O}}$ is performed as described above except as a function of varying H_2O and NO_x . The sensitivity is interrogated as a function of NO_x because the extent to which OH production occurs via primary ((R1) and (R2)) versus secondary (R3) routes determines how strongly $[\text{OH}]$ responds to variations in $[\text{H}_2\text{O}]$.

The $[\text{OH}]^{\text{TROP}}$ response to H_2O is found by calculating the relative change in H_2O on a monthly basis from both MERRA-2 and AIRS. Mixing ratios of H_2O from both data sets are averaged into 30° latitude bins, as shown in Figure S2. The MERRA-2 reanalysis suggests that H_2O in the lowermost troposphere increases with temperature at a rate of 7.3%/K globally and 5.4%/K in the tropics, roughly consistent with the Clausius-Clapeyron relation in which H_2O varies nonlinearly with temperature. The fractional change in H_2O , relative to 1980 for MERRA-2 and relative to the first year of available data spanning 2002–2003 for AIRS, is calculated on a monthly basis. The change in H_2O is then multiplied by the appropriate value of $C_{\text{H}_2\text{O}}$ (i.e., from the corresponding latitude, pressure, H_2O , and NO_x bin) to calculate a value of $\delta [\text{OH}]$ for each grid point in the base 3-D OH field obtained from GMI. The perturbed 3-D field of OH is then used to determine the change in $[\text{OH}]^{\text{TROP}}$ attributable to variations in H_2O . Because AIRS data are not available prior to 2002, the value of $[\text{OH}]^{\text{TROP}}$ calculated for the MERRA-2 reanalysis at year 2002 is treated as the baseline for subsequent AIRS $[\text{OH}]^{\text{TROP}}$ calculations. Our analysis of $[\text{OH}]^{\text{TROP}}$ focuses on relative differences from year to year, and all evaluations of trends in $[\text{OH}]^{\text{TROP}}$ due to H_2O are kept internally consistent (i.e., represent either the MERRA-2 from 1980 to 2015 or the AIRS trend from 2002 to 2015).

2.5.3. OH Response to Methane

The effect of methane on $[\text{OH}]^{\text{TROP}}$ is evaluated by using the DSMACC box model to estimate the feedback of methane on OH, denoted as C_{CH_4} , at varying levels of NO_x . Within each of the six latitude bins and four pressure bins identified in section 2.2, all tropospheric values of NO_x present in the GMI simulation are used to identify 10 NO_x bins. Mixing ratios of methane are varied in 10 steps spanning the observational record for our study period (1,640–1,850 ppb, determined from Dlugokencky et al., 2009). All other inputs to the box model are determined as averages of tropospheric values from GMI within each latitude and pressure bin. Output from these simulations are used to calculate C_{CH_4} as a function of methane and NO_x , in units of percent change in $[\text{OH}]$ per parts per billion change in methane. Each value of OH in the 3-D base field is then adjusted according to the value of C_{CH_4} for the appropriate latitude, pressure, and NO_x bin. Values of C_{CH_4} range from -0.39 for the cleanest background conditions (lowest methane and NO_x), up to -0.41 at high methane/low NO_x , and down to -0.08 at low methane/high NO_x . Values of C_{CH_4} encompass the estimate of the feedback of methane on tropospheric OH given by Prather et al. (2001), equivalent to $C_{\text{CH}_4} = -0.32$, while the values of C_{CH_4} of small magnitude at high NO_x indicate the role of secondary OH production via (R3). After adjusting the 3-D field of OH by the factor of C_{CH_4} as a function of NO_x , the $[\text{OH}]^{\text{TROP}}$ response to methane is calculated using Equation (1).

2.5.4. OH Response to Temperature

The effect of temperature variations on $[\text{OH}]^{\text{TROP}}$ is calculated to take into account both changes in chemical production of OH as well as its rate of reaction with methane. We perform box model calculations of $\delta (\text{OH})/\delta (T)$ as a function of $[\text{O}_3]$ following the same method of our methane sensitivity calculation, that is, probing conditions that span the full range of temperature and $[\text{O}_3]$ within the troposphere. This sensitivity is applied within the calculation of $[\text{OH}]^{\text{TROP}}$. Note that since the box model simulations are constrained by values of $[\text{H}_2\text{O}]$ determined for the latitude and pressure bins identified, the effect of temperature on $[\text{OH}]^{\text{TROP}}$ via water vapor is not evaluated here. Instead, that effect is encompassed by our evaluation of H_2O described in section 2.5.2. Simultaneously, temperature values used in the calculation of the rate constant for the $\text{OH} + \text{CH}_4$ reaction, or $k_{\text{OH} + \text{CH}_4}$ in equation (1), are allowed to vary based on year. As such, the total response of $[\text{OH}]^{\text{TROP}}$ to temperature encapsulates both changes in OH production and loss.

2.5.5. OH Response to NO_x

The changes in $[\text{OH}]^{\text{TROP}}$ that result from fluctuations in the global burden of NO_x are estimated using the 3-D NO_x field generated by the GMI global chemical transport model. While a recent study suggests that globally totaled NO_x emissions are almost constant for the period 2005–2014 (Miyazaki et al., 2017), analyses of NO_x

emissions on a regional basis from the same study and by Zhang et al. (2016) suggest significant shifts in the locations where emissions of NO_x have occurred. Therefore, variation in the tropospheric distribution of NO_x could be an important driver of changes in $[\text{OH}]^{\text{TROP}}$.

First, the interannual variations in NO_x emissions used within our GMI simulation are validated. The largest source of NO_x interannual variability in this simulation is biomass burning, which can be constrained utilizing satellite observations. Using the MODIS fire count data set described in section 2.4, the number of fire counts are totaled for the five biomass burning regions highlighted in Figure S3: (1) South America, (2) Equatorial Africa, (3) Southern Africa, (4) Siberia, and (5) Southeast Asia. The total NO biomass burning emissions are also found for each region (Figure S4). Figures S3 and S4 show the annual total fire counts and biomass burning NO emissions, respectively. However, correlations of the two quantities are performed by taking the totals on a monthly basis, then averaging across the biomass burning season for a given region and year. The correlations calculated by this analysis are shown in Figure S5. Overall, the NO emissions used in our GMI simulation capture the interannual effects of biomass burning variations; only the Southern Africa region yields a value of r^2 less than 0.65. van der Werf et al. (2006) identify significant mismatches in the timing of the South African biomass burning season when comparing their method, using MODIS fire data, to the methods utilized by other inventories, for example, CO or aerosol optical depth observations. Because the GFED2 database (van der Werf et al., 2006) is combined with the bottom-up RETRO inventory (Schultz et al., 2008) to derive the emissions used in our simulation of the GMI model, this discrepancy in timing is likely the driver of the lower correlation seen in Figure S5 for the South African region.

While we focus efforts to validate the NO_x emissions due to biomass burning within the GMI model, we note that variations in NO_x emissions due to other processes are also simulated. Variations in NO_x emissions from both lightning and soil sources are calculated online, as described in section 2.1, while anthropogenic emissions (for factors other than biomass burning) follow the MACCity inventory (Granier et al., 2011). While LNO_x emissions do vary within a given year, their total emissions per year are constrained to 6.5 Tg N/year. Figure S6 depicts the monthly emission rates of NO_x from biomass burning and lightning sources within the tropics for our study period. Figure S7 also shows NO_x emissions but totaled for the entire globe and including the additional sources of biofuels and fossil fuels. The fossil fuel source of NO_x is about 3–4 times the source of NO_x from lightning, but other work shows that emissions of LNO_x are more variable from year to year. While it was not possible to include observationally constrained LNO_x emissions in our simulation for the time period of interest (the first satellite observations of lightning flashes became available in 1995), we explore the possible implications for LNO_x and $[\text{OH}]^{\text{TROP}}$ through the following discussion of past research.

Previous studies indicate that $[\text{OH}]^{\text{TROP}}$ correlates strongly with LNO_x . Work presented in Fiore et al. (2006) suggests that rising emissions of LNO_x due to increases in convection are primarily responsible for an increase in $[\text{OH}]^{\text{TROP}}$ of +1.2% between 1991–1995 and 2000–2004. Even though the model simulations analyzed in Fiore et al. calculated substantially lower total emissions of LNO_x (maximizing at 2.7 Tg N/year by year 2004, compared to recent estimates as high as 9 Tg N/year, Nault et al., 2017), other studies also point to effects of lightning variability on $[\text{OH}]^{\text{TROP}}$. Murray et al. (2013) demonstrated that constraining lightning emissions using satellite observations of flashes, particularly in the tropics, improves model simulations in the interannual variability of $[\text{OH}]^{\text{TROP}}$. They found that emissions of NO_x from lightning in the tropics exhibit twice the variability when constrained to satellite flash observations, as compared to a control simulation in which flash rates vary with model-calculated cloud top heights. In addition, the influence of LNO_x emissions may extend down to the middle to lower troposphere, where impacts on OH chemistry are most relevant for the oxidation of methane (Murray, 2016). Many uncertainties still exist concerning the vertical distribution of flash frequency, cloud top heights, and the amount of NO_x produced per flash (Labrador et al., 2004). Regardless, our constraint to fixed annual emissions of LNO_x means that the response of $[\text{OH}]^{\text{TROP}}$ to changes in NO_x does not exhibit the variability that a full consideration of lightning would provide.

The perturbations to $[\text{OH}]^{\text{TROP}}$ due to NO_x are found, similarly to the other observational constraints, by using the DSMACC box model to calculate $\delta(\text{OH})/\delta(\text{NO}_x)$ as a function of HCHO, obtained from GMI. Formaldehyde is chosen due to its role as a proxy for alkyl reactivity as shown in (R4) and (R5). In other words, HCHO indicates the presence of RO_2 alkyl peroxy radicals that react with NO to regenerate OH, thus determining if NO_x will act as a sink or source of OH. While we do not separately analyze the role of the HNO_3 sink for the HO_x family of radicals (R8), our sensitivity analysis does account, for instances, at the highest NO_x

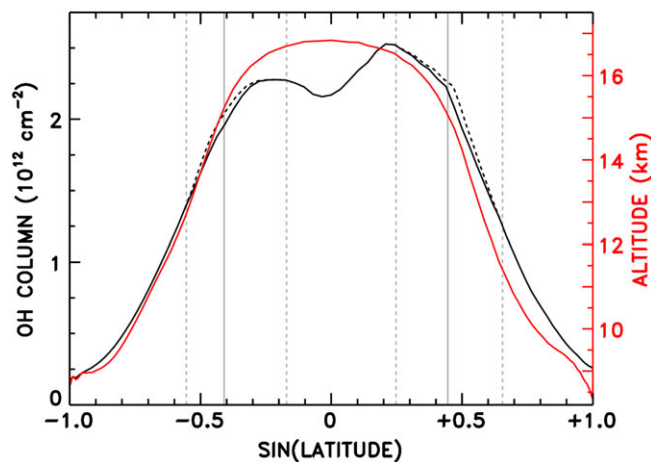


Figure 2. Annual zonal mean tropospheric OH column (black, solid) and tropopause altitude (red) are plotted against sin (latitude) from the Global Modeling Initiative model for year 1980. Solid gray lines show annual average latitudes of tropical boundaries. Dashed gray lines show minimum/maximum boundary latitudes, as a function of longitude, for each hemisphere. Note that while the metrics shown here are calculated for the entire year of 1980 from the GMI simulation, our analysis relies on these metrics found on a month-by-month basis. Black dashed lines show the simulated widening of the OH column distribution by 1.8° (rate of $0.5^\circ/\text{decade}$ over 36 years; see section 2.5.6 for further detail).

concentrations, when NO_x increases lead to decreases in OH concentrations. Roughly 10% of the sensitivities calculated, across all latitude, pressure, HCHO, and NO_x bins, exhibit values of $\delta(\text{OH})/\delta(\text{NO}_x)$ less than 0, reflecting the more important role of NO_x as a secondary production source of HO_x rather than as a sink (Lelieveld et al., 2016).

For each latitude and pressure bin described in section 2.2, NO_x and HCHO concentrations from GMI are divided into 10 bins spanning the range of tropospheric values. Using average values of all other parameters input as constraints to DSMACC (section 2.2) within each latitude/pressure bin, the box model is run to diurnal steady state and the variation in 24-hr average OH relative to the change in NO_x is calculated. Upon calculating the sensitivity of OH to changes in NO_x (C_{NO_x}), these values are combined with actual variations in NO_x (taken from GMI) to determine the perturbation to $[\text{OH}]^{\text{TROP}}$. Each value of OH in the 3-D GMI base field is adjusted according to the value of $\delta(\text{OH})/\delta(\text{NO}_x)$ for the appropriate HCHO bin and the corresponding change in NO_x , relative to year 1980, for all grid boxes. These changes in the 3-D OH field are propagated through to a new calculation of $[\text{OH}]^{\text{TROP}}$ following Equation (1), perturbed by the effect of changing NO_x .

2.5.6. OH Response to Tropical Expansion

To simulate the change in $[\text{OH}]^{\text{TROP}}$ that resulted from expansion of the tropics over the period of 1980–2015, we first have to determine the extent of the tropics in our initial 3-D field of OH from the GMI model.

We use the bimodal distribution of tropopause heights, first discussed in

Seidel and Randel (2007), to determine Hadley cell boundaries. While Seidel and Randel (2007) suggest a definition of tropical extent dependent on an absolute threshold in tropopause altitude, Davis and Rosenlof (2012) offer a more robust definition that instead uses relative changes in tropopause altitude. Following the Davis and Rosenlof (2012) definition, we identify the tropical boundary as the lowest latitude at which the tropopause altitude falls to 1.5 km below the mean tropopause altitude between 15°S and 15°N . We tested several additional definitions of tropical extent but found that averaging over the relatively short time span of 1 model year (from 12 monthly averages) was insufficient for producing smooth, unbroken tropical boundaries for all but the tropopause altitude metric. We determine tropical boundaries using tropopause height for each longitude, in both the NH and SH, for each month of the GMI base simulation. The tropical boundaries that we identify are shown as black lines in Figure 1.

The primary mechanism through which we expect tropical widening to affect $[\text{OH}]^{\text{TROP}}$ is primarily through the change in overhead O_3 , which in turn follows variations in the tropopause height (Hudson et al., 2003). Increases in H_2O that would accompany the transition of a location from an extratropical to a tropical regime would also tend to increase OH concentrations. The correlation of OH with tropopause height is clear when analyzing zonal mean values of tropospheric OH column as output by the GMI model (Figure 2). The values depicted in Figure 2 are also annually averaged for illustration purposes, though we note that analysis of simulated tropical widening is performed on a monthly basis. The Pearson correlation coefficient of annually, zonally averaged OH column versus tropopause altitude is 0.99. Without averaging, the correlation coefficient of the longitude dependent, monthly values is 0.85 due to zonal asymmetries in the OH column that are driven by factors other than tropopause altitude and overhead O_3 (such as NO_x). We impose the effects of tropical expansion by essentially widening the latitudinal distribution of tropospheric OH column values, seen as the black dotted line in Figures 2 and S8, which shows a zoomed-in view of the NH. This is done by placing anchor points at the minimum and maximum latitudes identified as tropical boundaries, as a function of longitude, for a given hemisphere and month. The OH column value at the mean tropical boundary latitude is shifted poleward by the number of degrees of tropical widening being imposed. Here we use the rates of $0.35^\circ/\text{decade}$ in the NH and $0.17^\circ/\text{decade}$ in the SH as estimated from observations by Allen et al. (2014) to simulate the rate of Hadley cell expansion and approximate the resulting sensitivity of $[\text{OH}]^{\text{TROP}}$. We also use values of $0.5^\circ/\text{decade}$ in each hemisphere to represent the maximum rate, reported by Davis and Rosenlof (2012) through examination of mean meridional streamfunction, and of $0.05^\circ/\text{decade}$ in the NH,

0.13° decade $^{-1}$ in the SH to represent the rate of widening simulated by models that participated in the Coupled Model Intercomparison Project Phase 5 (CMIP5) (Allen et al., 2014). The poleward shift of tropospheric OH columns maximizes at the mean tropical boundary latitude and is interpolated to a shift of 0° at the anchor points. We calculate the fractional change in the tropospheric OH column at each latitude, depicted in Figure S9, using this new distribution (black dashed line, Figures 2 and S8), and apply this within our calculation of $[\text{OH}]^{\text{TROP}}$.

We consider the effects of tropical widening and changes in overhead column O_3 on $[\text{OH}]^{\text{TROP}}$ as independent factors in this analysis. The justification for doing so is that the primary effect of tropical widening on $[\text{OH}]^{\text{TROP}}$ is driven by the exposure of more regions of the troposphere to climatological tropical conditions (low column O_3 , high humidity), rather than a variation of either column O_3 or humidity within the latitudes defined to be the tropics at any point in time. In other words, the third and fourth latitude bins (30°S – 0° ; 0° – 30°N section 2.2) used in our analysis should be expanded to account for the widening of the tropics, whereas the second and fifth bins (60°S – 30°S ; 30°N – 60°N) should be contracted to compensate. Furthermore, the observed changes in total O_3 result from factors not related to tropical expansion, including the decreasing atmospheric burden of halogenated compounds (Butchart & Scaife, 2001; Li et al., 2009) and the injection of volcanic sulfate aerosols into the stratosphere (Fahey et al., 1993), both of which impart larger changes in overhead O_3 than are expected as a result of tropical expansion. Additionally, increases in H_2O are expected to accompany the decreases in overhead O_3 as a location transitions to a tropical regime. Given the multiple factors that likely affect $[\text{OH}]^{\text{TROP}}$ during the process of tropical widening and the low level of agreement between metrics of Hadley cell expansion (Davis & Rosenlof, 2012), we choose to approximate the effect by estimating the impact on $[\text{OH}]^{\text{TROP}}$ of changing the boundaries of our model boxes, albeit in simplified fashion. It is possible that other complex processes, such as an increasing magnitude of stratosphere-troposphere ozone flux as a result of climate change (Hegglin & Shepherd, 2009), also influence $[\text{OH}]^{\text{TROP}}$, though we have not sought to examine those here. Ultimately, improving the dynamics within global models such that tropical expansion and other changes in circulation are simulated with accuracy will enable a more process-oriented approach to this question.

3. Results and Discussion

The results of our $[\text{OH}]^{\text{TROP}}$ analysis are shown in Figure 3. Figure 3a shows the evolution of methane observations from 1983 to 2015, with mixing ratios prior to 1983 filled in via interpolation as described in section 2.4. Figure 3b shows our calculation of $[\text{OH}]^{\text{TROP}}$, with the initial 1980 value of $1.241 \times 10^6 \text{ cm}^{-3}$ from the base field of OH from GMI. This value of $[\text{OH}]^{\text{TROP}}$ gives a lifetime for methane with respect to reaction with OH of 9.8 years, which is lower than that inferred from MCF (11.2 years, estimated by Prather et al., 2012, for year 2010) but within the range of global model variability described in Table 1 of Voulgarakis et al. (2013) for year 2000. While observation-based estimates of methane lifetime do not exist for the year of our simulation, Naik et al. (2013) provide a multimodel mean value for year 1980 from the Atmospheric Chemistry and Climate Model Intercomparison Project, 10.2 ± 1.7 years, that encompasses our value from GMI.

The time varying responses of $[\text{OH}]^{\text{TROP}}$ to column O_3 , MERRA-2 H_2O , AIRS H_2O , methane, NO_x , and temperature are shown as dark green, purple, orange, red, brown, and light green lines, respectively, while the trend in $[\text{OH}]^{\text{TROP}}$ determined as a result of increases in tropical width is shown as a blue line. The sum of all six responses, with only the MERRA-2 source of H_2O included due to its coverage of the entire period, gives the total change in $[\text{OH}]^{\text{TROP}}$ shown in Figure 3c.

Our calculations show the putative expected decline in $[\text{OH}]^{\text{TROP}}$ due to rising methane (Prather et al., 2001) is buffered by the other factors. The slope of the best fit line to the total change in $[\text{OH}]^{\text{TROP}}$ (Figure 3c) is $-0.08 \pm 0.19\%$ /decade, while the trend due to methane alone is $-1.01 \pm 0.05\%$ /decade. Here and throughout, the given uncertainties represent 1-sigma uncertainties in the calculated slope. Linearly regressing the contributions of the individual factors results in $[\text{OH}]^{\text{TROP}}$ trends, in order of importance, of $+0.44 \pm 0.20\%$ /decade due to rising H_2O , $+0.25 \pm 0.07\%$ /decade due to rising NO_x , $+0.13 \pm 0.11\%$ /decade due to variations in O_3 column, $+0.12^{+0.07}_{-0.09}\%$ /decade due to tropical widening, and $-0.02 \pm 0.02\%$ /decade due to increasing temperature. Totalling the individual trends in $[\text{OH}]^{\text{TROP}}$ listed here, plus the trend due to methane, gives a

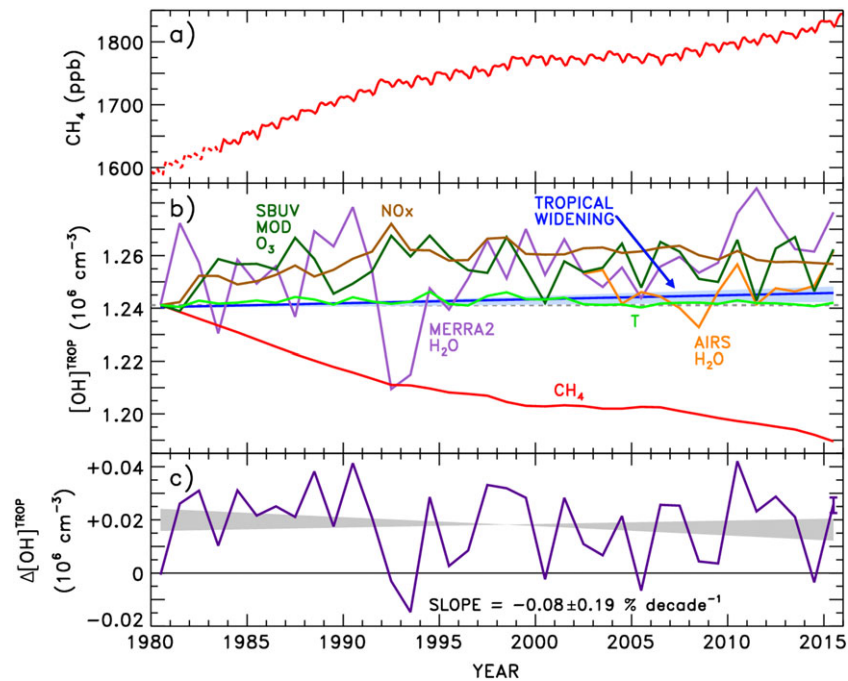


Figure 3. The monthly evolution of surface methane for 1980–2015 is shown in panel (a). Dashed red line indicates the period for which data are not available and methane is inferred by linearly interpolating backward using the trend from 1983–1990 on a month-by-month basis. Panel (b) shows annually averaged $[\text{OH}]^{\text{TROP}}$ variations attributable to changes in column O_3 (dark green line), H_2O (obtained from the Modern-Era Retrospective Analysis for Research and Applications Version 2, MERRA-2, reanalysis, purple, and the Atmospheric Infrared Sounder, AIRS, satellite instrument, orange), methane (red), temperature (light green), and widening of the tropics (blue) at a rate of $0.52^\circ/\text{decade}$ globally. The light blue shading represents a range of possible $[\text{OH}]^{\text{TROP}}$ values using lower and upper limits of the tropical expansion rate (0.18° and $1.0^\circ/\text{decade}$, respectively). Panel (c) shows the total change in $[\text{OH}]^{\text{TROP}}$, calculated as the sum of the change in $[\text{OH}]^{\text{TROP}}$ from the six factors shown in panel (b) (overhead O_3 , H_2O , methane, NO_x , T , and tropical widening, with H_2O taken from MERRA-2) relative to 1980. The gray shaded region depicts the 1-sigma uncertainty in the slope of the best fit line calculated for the net change in $[\text{OH}]^{\text{TROP}}$.

total value of $-0.09 \pm 0.26\%/ \text{decade}$, where the uncertainty is calculated as the root sum of squares. This is similar to the trend derived by performing a linear fit of the total anomaly (Figure 3c), which yields the $-0.08 \pm 0.19\%/ \text{decade}$ given above. Both of these values (i.e., $-0.09 \pm 0.26\%/ \text{decade}$ and $-0.08 \pm 0.19\%/ \text{decade}$) include uncertainties that span zero, making any trend in $[\text{OH}]^{\text{TROP}}$ indistinguishable in this analysis. Therefore, increases in $[\text{OH}]^{\text{TROP}}$ arising from the combined influence of H_2O , NO_x , overhead O_3 , and tropical widening may have countered practically all of the expected decrease in $[\text{OH}]^{\text{TROP}}$ due to rising methane. The value for tropical widening stated in this paragraph is based on the $0.35^\circ/\text{decade}$ (NH) and $0.17^\circ/\text{decade}$ (SH) estimates from Allen et al. (2014); we include further discussion of the $[\text{OH}]^{\text{TROP}}$ trend attributed to tropical widening below.

The magnitude of the interannual variability of $[\text{OH}]^{\text{TROP}}$ shown in Figure 3c agrees with the relatively small estimate of $\sim 2\%$ calculated by Montzka et al. (2011) for the years 1998–2008. To further compare the variability in $[\text{OH}]^{\text{TROP}}$, we calculate with previous studies, Figure 4 shows annual percent anomalies from our results as well as from Montzka et al. (2011), McNorton et al. (2016), Rigby et al. (2017), and Turner et al. (2017). Our net trend in $[\text{OH}]^{\text{TROP}}$ is not in full qualitative agreement with these past studies that derive $[\text{OH}]^{\text{TROP}}$ from MCF observations. While we calculate small fluctuations in the anomaly of $[\text{OH}]^{\text{TROP}}$ between about 0% and 2% during the stall in methane growth, Turner et al. (2017), for instance, find an increase in $[\text{OH}]^{\text{TROP}}$ of 7% from 1991 to 2000 followed by a decrease of 7% from 2003 to 2016. Our result indicates that OH variations may not have had as large a role in causing the slowdown in methane growth rate as is suggested by other studies.

We also show in Figure 4 the evolution in $[\text{OH}]^{\text{TROP}}$ from the full GMI simulation (green line), which gives some indication of how much influence we may expect from additional factors for which we did not account. For

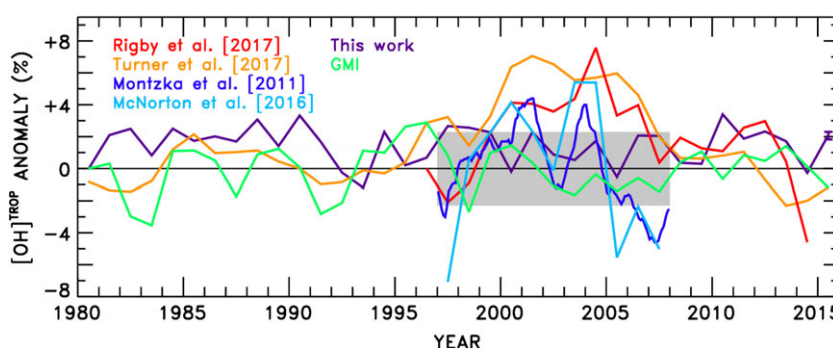


Figure 4. Time series of $[\text{OH}]^{\text{TROP}}$ percent anomalies from our empirical analysis (dark purple) shown with $[\text{OH}]^{\text{TROP}}$ anomalies from previous studies. Results of Rigby et al. (2017) are shown in red, from Turner et al. (2017) in orange, from Montzka et al. (2011) in dark blue, and from McNorton et al. (2016) in light blue. In cases of past studies that used both National Oceanic and Atmospheric Administration (NOAA) and Advanced Global Atmospheric Gases Experiment (AGAGE) networks of methyl chloroform observations to evaluate $[\text{OH}]^{\text{TROP}}$, we show only results from the NOAA measurements. The mean interannual variability in $[\text{OH}]^{\text{TROP}}$ found by Montzka et al. (2011) for the period 1997–2007 is denoted by the gray shaded region ($\pm 2.3\%$). We also calculate the $[\text{OH}]^{\text{TROP}}$ anomaly simulated by our Global Modeling Initiative (GMI) chemical transport model (orange line). Anomalies in $[\text{OH}]^{\text{TROP}}$ from this work, and from GMI are calculated relative to year 1980. Rigby et al. anomalies are calculated relative to the initial value of $[\text{OH}]^{\text{TROP}}$ representative of year 1996.

instance, coupling and secondary effects of the processes we investigate (e.g., the presumed increase in tropospheric O_3 that would accompany the increase in methane concentrations) and responses to parameters without sufficient observational constraint to include here (e.g., VOCs) are encompassed in the GMI calculation of $[\text{OH}]^{\text{TROP}}$. While the anomalies in $[\text{OH}]^{\text{TROP}}$ from GMI show variations not fully synchronized with our empirical anomalies, there are similarities, and interannual variability from our empirical estimates (1.6%) is close in magnitude to that from GMI (1.3%). It is important to remember that GMI is not tightly constrained to observations, though, so there is little reason to expect GMI to be a robust metric of accuracy. Interestingly, while the empirically derived anomalies in $[\text{OH}]^{\text{TROP}}$ do not capture the increase in the early 2000s suggested by the chemical inversion studies (McNorton et al., 2016; Montzka et al., 2011; Rigby et al., 2017; Turner et al., 2017), neither does the GMI simulation, perhaps suggesting that full accounting of the sources of OH as well as sinks causes $[\text{OH}]^{\text{TROP}}$ to be well buffered over time.

The response of $[\text{OH}]^{\text{TROP}}$ to H_2O reported above ($+0.44 \pm 0.20\%$) refers only to the trend calculated using the MERRA-2 reanalysis fields for the entire 1980–2015 period. The rise in $[\text{OH}]^{\text{TROP}}$ derived from satellite observations of H_2O , that is, the orange line in Figure 3b, is of a similar magnitude albeit higher uncertainty: $+0.50 \pm 0.45\%/decade$. Although the period over which we have observational constraint on tropospheric H_2O is too short to establish a robust long-term trend in $[\text{OH}]^{\text{TROP}}$, the data record is consistent with the magnitude of change we calculated from reanalysis fields of H_2O .

The factor imparting the next largest trend in $[\text{OH}]^{\text{TROP}}$ is variations in NO_x . The increase in $[\text{OH}]^{\text{TROP}}$ due to NO_x is $+0.25 \pm 0.07\%/decade$, resulting from increases in the NO_x burden and subsequently enhanced secondary production of OH via reactions (R3) and (R4). The growing burden of NO_x occurs mostly due to the fossil fuel emissions inventory, with the largest growth occurring between 1980 and 1990 (Figure S7). The response of $[\text{OH}]^{\text{TROP}}$ follows suit, with most of the increase due to NO_x (brown line, Figure 3) occurring prior to 1993. Values of $[\text{OH}]^{\text{TROP}}$ due to NO_x level off after this point and decline slightly, due to the decrease in fossil fuel emissions between 1990 and 2000. We again note that since LNO_x emissions in the GMI model are constrained to an annual total of 6.5 Tg N/year as discussed in section 2.5.5, the NO_x variations we rely on for our analysis do not represent interannual changes in the lightning source. As a result, the variation in $[\text{OH}]^{\text{TROP}}$ due to NO_x from 1 year to the next is likely underestimated in our framework. However, empirically based estimates of the time variation of LNO_x are not available for much of our time period of interest.

Our analysis indicates that overhead O_3 is the third largest influence on $[\text{OH}]^{\text{TROP}}$ between 1980 and 2015. The increase in $[\text{OH}]^{\text{TROP}}$ of $+0.13 \pm 0.11\%/decade$ due to O_3 has a relatively large uncertainty, as was the case with H_2O from both observation-based data sets. The response of $[\text{OH}]^{\text{TROP}}$ to both H_2O and column O_3

Table 1

Percent Changes in Methane Lifetime Due to Reaction With OH (τ_{CH_4+OH}) Identified by Holmes et al. (2013) and in $[OH]^{TROP}$ Identified by This Study Between 1980–1985 and 2000–2005

Parameter	$(\Delta\tau_{CH_4+OH} \text{ from Holmes et al., 2013}) \times -1$ (%)	$\Delta [OH]^{TROP}$ from this study (%)
CH ₄	-3.18 ± 0.50	-2.60 ± 0.40
T	$+0.34 \pm 0.09$	$+0.01 \pm 0.14$
H ₂ O	$+1.74 \pm 0.16$	$+0.14 \pm 1.32$
Overhead O ₃	$+0.68 \pm 0.14$	$+0.27 \pm 0.93$
NO _x	$+2.85 \pm 1.69^a$	$+1.09 \pm 0.41^b$
Hadley cell expansion	—	$+0.25^{+0.13}_{-0.19}^c$
Total trend	$+2.37 \pm 1.74$	-0.80 ± 1.47

^aThe NO_x response reported here represents combined responses of NO_x from anthropogenic, shipping, aviation, biomass burning, and lightning shown in Holmes et al. with uncertainties calculated as the root sum of squares. ^bThe NO_x response reported from our study represents all NO_x sources. ^cThe uncertainties in $\Delta [OH]^{TROP}$ attributed to Hadley cell expansion represent the maximum and minimum rates of tropical widening tested here; refer to section 2.5.6 for further detail.

shown in Figure 3b indicates significant year-to-year variability, rather than a monotonic trend. Variations in overhead O₃ are driven to an extent by the sources of tropospheric O₃, which are in turn affected by biomass burning, El Niño/La Niña conditions, and lightning (Chandra et al., 2007; Inness et al., 2015; Logan et al., 2008; Ziemke & Chandra, 2003). The bulk of O₃ in the stratosphere, however, is sensitive to the burden of anthropogenic halocarbons (Butchart & Scaife, 2001; Li et al., 2009) and the injection of volcanic sulfate aerosols into the stratosphere (Fahey et al., 1993). The signal in column O₃ due to the eruption of Mount Pinatubo, however, is discernible in both the O₃ observations (depressed O₃ column values in 1992–1994 particularly in the southern tropics and midlatitudes of Figure S1) and in the $[OH]^{TROP}$ response to overhead O₃ (highest values present at years 1992 and 1994, green line, Figure 3b). During the same period, tropospheric H₂O also declined due to the temperature decrease caused by the Pinatubo eruption (Randel et al., 1996; Soden et al., 2002), driving $[OH]^{TROP}$ down and overwhelming the effect of overhead O₃, resulting in a net drop in $[OH]^{TROP}$ (Figure 3c). In addition, $[OH]^{TROP}$ likely responded to the attenuation in UV flux that occurred due to absorption by SO₂ and aerosol extinction (Bändä et al., 2015; Dlugokencky et al., 1996), but our results do not capture effects of UV variation due to sources

besides overhead O₃. Consequently, a further decrease in $[OH]^{TROP}$ due to the eruption of Pinatubo probably occurred. Since the attenuation of UV light by volcanic gases and aerosols is short lived and occurred near the middle of the period of interest, the effect on long-term trends in $[OH]^{TROP}$ is likely small. Overall, overhead O₃ leads to a rise in $[OH]^{TROP}$ because the thickness of the global ozone layer had declined over most of the period of study (WMO, 2014). The halogen loading of the midlatitude, lower stratosphere peaked between 1996 and 1999 and is now slowing declining, resulting in a small steady rise in overhead O₃ since about 2000 (Chipperfield et al., 2017). As a result, future changes in $[OH]^{TROP}$ driven by overhead O₃ will likely reinforce the methane-induced decline in this quantity.

The negligible influence of temperature on $[OH]^{TROP}$ contrasts to results from past studies. Notably, Holmes et al. (2013) identified the strongest sensitivity of the lifetime of methane lifetime to this factor. However, while the sensitivity is large, the total change in the lifetime of methane between 1980 and 2005 found by Holmes et al. is relatively small compared to other factors that were quantified here, including H₂O, anthropogenic NO_x emissions, and the influence of methane. Since the midtropical troposphere (i.e., the region of the atmosphere with a temperature close to 272 K) exerts primary control on the loss of methane and thus on $[OH]^{TROP}$, the small value we compute for the impact of temperature on tropospheric oxidizing capacity is not surprising. Numerous observations show little warming in the midtropical troposphere: for example, Figure 5.6 of Santer et al. (2018) shows trends in temperature hovering near zero at 500 hPa, 20°S to 20°N, between 1979 and 2004. The expression of global warming in the tropics is more apparent in rising temperature near the surface and in the upper troposphere than in the midtroposphere, plus of course possibly driving the expansion of the tropics.

Further comparison to the results of Holmes et al. (2013) reveals some large differences and a few similarities. Table 1 compares the trends in methane lifetime attributed to various factors by Holmes et al. for 1980–2005 (multiplied by -1 to account for the inverse relationship between methane lifetime and OH burden) with trends in $[OH]^{TROP}$ we evaluate for the same period. The trends are calculated as the percent change in methane lifetime or $[OH]^{TROP}$ between the average for the period 1980–1985 and the average for 2000–2005, to minimize the role of interannual variability. Note that the units used in Table 1 differ from the units used elsewhere in this paper for two reasons: (1) values in Table 1 are reported as total percentage changes over a time period (%) rather than a rate of change (%/decade) to match the units used by Holmes et al. (2013) and (2) the time period over which values in Table 1 are evaluated (1980–2005) is shorter than the time period over which our primary analysis is conducted (1980–2015) again to allow for a comparison to the results of Holmes et al. (2013). Responses due to the rise in methane are similar between the studies, $-3.18 \pm 0.50\%$ from Holmes et al. and $-2.60 \pm 0.40\%$ from our analysis. The trend in $[OH]^{TROP}$ resulting

from H₂O in Holmes et al. ($+1.74 \pm 0.16\%$) differs markedly from our value ($+0.14 \pm 1.32\%$), as does the trend due to temperature ($+0.34 \pm 0.09\%$ from Holmes et al. and $+0.01 \pm 0.14\%$ from our analysis). As pointed out by Holmes et al., water vapor trends differ among reanalysis data sets (Trenberth et al., 2011). So despite both studies using MERRA products, our use of MERRA version 2, with noted improvements to its representation of the water cycle (Gelaro et al., 2017), may explain the difference in the effect of H₂O on methane obtained by Holmes et al. based upon MERRA version 1. Indeed, Gelaro et al. (2017) show significant differences in specific humidity between MERRA and MERRA-2 (in their Figure 8), and cursory analysis suggests generally smaller increasing trends in specific humidity within MERRA-2 compared to MERRA. Trends in $[\text{OH}]^{\text{TROP}}$ attributed to overhead O₃ differ by more than a factor of two; Holmes et al. report $+0.68 \pm 0.14\%$, while we find $+0.27 \pm 0.93\%$. Responses due to NO_x are similarly large ($+2.85 \pm 1.69\%$ from Holmes et al. and $+1.09 \pm 0.41\%$ here) though the value from Holmes et al. combines responses that separately consider changes in anthropogenic, shipping, aviation, biomass burning, and lightning NO_x sources. Our value represents simultaneous simulation of changes in all NO_x sources. Finally, the overall trend in $[\text{OH}]^{\text{TROP}}$ identified in the work of Holmes et al. differs markedly from the trend we evaluate between 1980–1985 and 2000–2005: Holmes et al. report $+2.37 \pm 1.74\%$; we calculate $-0.80 \pm 1.47\%$. The difference in sign in the overall trend, despite our similar values for the portion of the trend attributable to methane, results from the compensating factors that impart positive trends in $[\text{OH}]^{\text{TROP}}$. In other words, Holmes et al. find larger increases in $[\text{OH}]^{\text{TROP}}$ due to H₂O, temperature, NO_x, and O₃ column than we do, accounting for most of the discrepancy in our total trends. While Holmes et al. cite several studies that similarly conclude $[\text{OH}]^{\text{TROP}}$ has increased from 1980 to 2005 (Dentener et al., 2003; Duncan et al., 2000; Hess & Mahowald, 2009; John et al., 2012; Karlsdottir & Isaksen, 2000; Naik et al., 2013; Stevenson et al., 2005), the recent study by Turner et al. (2017) points out that this finding relies on assumptions regarding methane emissions. When a sensitivity test using constant methane emissions over time is conducted, $[\text{OH}]^{\text{TROP}}$ remains nearly constant between 1980 and 2005, as seen in Figure 5 of Turner et al. While neither Turner et al. nor we posit that methane emissions are truly constant, we suggest that observational constraints on methane emissions are not sufficient to differentiate between slight increases and decreases in the OH burden during this period (i.e., 1980–2005).

Other qualitative comparisons between our work and that of Holmes et al. lend support to our empirical model of temporal variations in the tropospheric OH burden. Holmes et al. note years in which methane lifetime increases due to La Niña conditions (1989 and 1999–2000) and decreases due to El Niño conditions (1982–1983, 1987–1988, and 1997–1998). These increases and decreases in methane lifetime correspond well to the decreases and increases, respectively, we see in $[\text{OH}]^{\text{TROP}}$. These responses are driven by a combination of NO_x, H₂O, and overhead O₃, which vary primarily as a result of differences in convection and locations of biomass burning events (Logan et al., 2008).

The effects of tropical widening on OH for the evaluated rates of expansion are modest. The current best estimate of the rate of tropical expansion based on observations from Allen et al. (2014) yields a $+0.12^{+0.07}_{-0.09}\%$ /decade trend in $[\text{OH}]^{\text{TROP}}$ between 1980 and 2015. The uncertainty reported for this value represents the high and low estimates of the rate of tropical expansion. The rate of decrease in $[\text{OH}]^{\text{TROP}}$ from increasing levels of methane is -1.01% /decade, so tropical widening may have offset $\sim 12\%$ of the effect of methane on the global abundance of OH. If the rate of Hadley cell expansion lies closer to 1° /decade globally, as suggested by Davis and Rosenlof (2012), the trend in $[\text{OH}]^{\text{TROP}}$ due to tropical widening lies at the high end of our uncertainty range, $+0.19\%$ /decade. The low end of our uncertainty estimate represents the rate of tropical expansion simulated by global models (Allen et al., 2014), resulting in a $[\text{OH}]^{\text{TROP}}$ trend of $+0.03\%$ /decade. Consequently, global models may underestimate the role of Hadley cell expansion in influencing $[\text{OH}]^{\text{TROP}}$ by about a factor of 4 to 6 (i.e., $0.12/0.03$ to $0.19/0.03$).

Because we did not quantify the impact of tropical widening on $[\text{OH}]^{\text{TROP}}$ using a process-based model, we necessarily cannot claim high accuracy in the quoted numbers in the paragraph above. Rather, we suggest that these be interpreted as a first approximation of the role of this impact. The difference in magnitude of $\Delta [\text{OH}]^{\text{TROP}}$ between the various estimates of the rate of tropical expansion, though, likely can be assigned a higher level of confidence, and the fact remains that current free-running models are missing an important driver of the oxidizing capacity of the global troposphere. Also, without considering the many factors that influence local concentrations of OH (e.g., CO, VOCs, and aerosols), we do not claim to have fully accounted for the budget of $[\text{OH}]^{\text{TROP}}$ through our analysis. The results of Holmes et al. (2013) suggests we have included

the major drivers of variability in $[\text{OH}]^{\text{TROP}}$, though the recent work of Gaubert et al. (2017) indicates a role for variations in CO for the most recent decade.

While the causes of the expansion of the Hadley circulation are still uncertain, recent studies are beginning to untangle the mechanisms behind the observed shifts in tropical boundaries. Early studies attributed tropical widening to cooling of the stratosphere caused by increasing greenhouse gases and stratospheric O_3 depletion (Lu et al., 2009; Seidel & Randel, 2006). Later, the drivers of tropical widening were shown to be more hemispheric in nature: formation of the O_3 hole in late austral spring is widely regarded as the driver of the SH tropical expansion (Kang et al., 2011; Polvani et al., 2011; Son et al., 2009, 2010), while Allen et al. (2012) found tropospheric O_3 and black carbon to be likely drivers of NH Hadley cell widening. Further, Allen et al. (2014) suggested that NH tropical width is influenced by sea surface temperatures, which are in turn forced by aerosols. If these studies are correct, then as stratospheric O_3 recovers and developed nations continue striving to improve air quality by reducing emissions of aerosols (Hoesly et al., 2018), it is possible that the rate of tropical widening will slow. Most recently, studies indicate that the observed changes in the Hadley circulation may result from unforced internal variability instead of a forced response (Allen & Kovilakam, 2017; Garfinkel et al., 2015). If further expansion of the Hadley cell fails to manifest, then the imperative to include this effect when studying future changes in atmospheric composition may safely be ignored. However, if tropical widening is in fact a forced response to climate change or other factors and continues to occur, then this factor will continue to partially offset future declines in $[\text{OH}]^{\text{TROP}}$ driven by the presumed continued rise in tropospheric methane.

4. Conclusions

This study uses observations of total column O_3 , H_2O , temperature, and methane along with simulated NO_x and an approximation of the effect of the widening of the Hadley cell to calculate variations in $[\text{OH}]^{\text{TROP}}$ for 1980 to 2015. Here $[\text{OH}]^{\text{TROP}}$ refers to the burden of OH in the global troposphere appropriate for calculating the lifetime of methane (section 2.3). The total trend in $[\text{OH}]^{\text{TROP}}$ identified from these six factors is $-0.08 \pm 0.19\%$ /decade relative to the annual average baseline value of $1.241 \times 10^6 \text{ cm}^{-3}$ for the tropospheric OH burden. While increases in atmospheric methane alone should have caused a theoretical average rate of decrease in $[\text{OH}]^{\text{TROP}}$ of $1.01 \pm 0.05\%$ /decade, the combined effects of overhead O_3 , H_2O , NO_x , temperature, and tropical widening limited this expected decrease in $[\text{OH}]^{\text{TROP}}$ to only 0.08% /decade. In order of importance, factors other than methane that drive the oxidizing capacity of the global troposphere are estimated to have caused changes in $[\text{OH}]^{\text{TROP}}$ of $+0.44 \pm 0.20\%$ /decade (H_2O), $+0.25 \pm 0.07\%$ /decade (NO_x), $+0.13 \pm 0.11\%$ /decade (overhead column O_3), $+0.12 (+0.07, -0.09)\%$ /decade (tropical widening), and $-0.02 \pm 0.02\%$ /decade (temperature). The effect of H_2O on tropospheric OH is driven by the Clausius-Clapeyron rise in humidity, due to global warming over 1980 to 2015. The NO_x -induced increase in OH is caused by rising anthropogenic emissions of NO_x , primarily due to combustion of fossil fuels, followed by secondary production of OH due to the reaction of HO_2 and various RO_2 species with NO. The increase in OH due to declining overhead column O_3 is caused by anthropogenic emissions of halocarbons that deplete stratospheric ozone. Since the global ozone layer has begun to recover, albeit slowly, this factor is expected to reverse sign. The cause of tropical widening is still the subject of active research. While the influence of tropical widening on the tropospheric OH burden is currently estimated to be modest, this phenomenon could have offset nearly 20% of the methane-induced decline in $[\text{OH}]^{\text{TROP}}$. Tropical widening is worth monitoring in the future, especially because free-running models tend to underestimate the extent of tropical widening and thus the impact on $[\text{OH}]^{\text{TROP}}$ by a considerable amount (i.e., a factor of 4 to 6). Finally, the effect of rising temperature on $[\text{OH}]^{\text{TROP}}$ is small, because the crucial region for loss of methane (the midtropical troposphere) has showed little evidence of warming, compared to more apparent warming at the surface and in the upper troposphere (Santer et al., 2018).

These findings differ from the results of a prior study by Holmes et al. (2013) that found larger increases in $[\text{OH}]^{\text{TROP}}$ due to temperature, H_2O , overhead O_3 , and NO_x . The reasons for differences in the $[\text{OH}]^{\text{TROP}}$ responses to temperature and H_2O may in part be due to our use of the updated MERRA-2 reanalysis data set, which includes significant improvements, particularly in specific humidity, compared to MERRA version 1, used to constrain the Holmes et al. simulations (Gelaro et al., 2017). The causes of differences in the $[\text{OH}]^{\text{TROP}}$ responses to NO_x and overhead O_3 variations between our analysis and Holmes et al.,

though, are less clear. Holmes et al. used a global model with fully interactive tropospheric chemistry, meaning that coupling between the processes affecting OH chemistry may be better simulated than in our empirical approach. However, a simulation of the GMI model analyzed here (green line, Figure 4) exhibits variations in $[\text{OH}]^{\text{TROP}}$ that are similar in magnitude to those that we calculate empirically. The total increase in $[\text{OH}]^{\text{TROP}}$ of $+2.37 \pm 1.74\%$ between 1980 and 2005 found by Holmes et al. is not supported by either our empirical analysis or the GMI chemical transport model.

Our results show low interannual variability in $[\text{OH}]^{\text{TROP}}$ (1.6% mean anomaly), which is comparable to previous estimates derived from observations of CH_3CCl_3 (MCF; 2.3% from Montzka et al., 2011). This supports the notion that $[\text{OH}]^{\text{TROP}}$ is well buffered in the atmosphere, as opposed to earlier work by Bousquet et al. (2005) and Prinn et al. (2005) that found year-to-year fluctuations in $[\text{OH}]^{\text{TROP}}$ on the order of 10%. Our calculations also do not support a discernable rise and fall in $[\text{OH}]^{\text{TROP}}$ corresponding to the pause in growth of atmospheric methane concentrations, in contrast to analyses of $[\text{OH}]^{\text{TROP}}$ inferred from MCF by Montzka et al. (2011), McNorton et al. (2016), Rigby et al. (2017), and Turner et al. (2017). Turner et al. calculated a rise and subsequent fall in $[\text{OH}]^{\text{TROP}}$ of 7% centered about the early 2000s, which they suggested contributed to the stall in the growth of tropospheric methane. Rigby et al. also found a rise in $[\text{OH}]^{\text{TROP}}$ of $\sim 10\%$ prior to 2004. These studies compute $[\text{OH}]^{\text{TROP}}$ based upon measurements of the temporal evolution of the atmospheric abundance of MCF and the supply of MCF to the atmosphere from industrial sources, as well as (for some studies) a few additional observations such as methane and its isotopes. Some of these studies use sophisticated three-dimensional chemical transport models to infer $[\text{OH}]^{\text{TROP}}$ from MCF; none of these studies address the fundamental reason why inferred $[\text{OH}]^{\text{TROP}}$ is varying (Prather & Holmes, 2017), and all are subject to uncertainties in MCF emissions both from industry as well as release from the ocean (Wennberg et al., 2004). By including the temporal variation of factors that govern both the sources and sinks of tropospheric OH, our analysis is designed to fully account for the production and loss of OH. While the MCF-based inferences of $[\text{OH}]^{\text{TROP}}$ provide important empirical constraints on the oxidizing capacity of the global troposphere, complete understanding of the variability and long-term trends of $[\text{OH}]^{\text{TROP}}$ will only be gleaned through consideration of the many factors that influence the chemistry of OH.

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